EE3032 Introduction to VLSI Design—Final

1. (26%) Answer the following problems.
   (a) List major steps of the full-custom design flow. (8%)
   (b) List two basic rules for drawing the layout of a standard cell. (4%)
   (c) List two major problems of dynamic logic. (4%)
   (d) List two limitations of CMOS domino logic. (4%)
   (e) What are core-limited design and pad-limited design? (6%)

2. (10%) What logic function does the circuit of Fig. 1 realize? Explain your answer in detail.

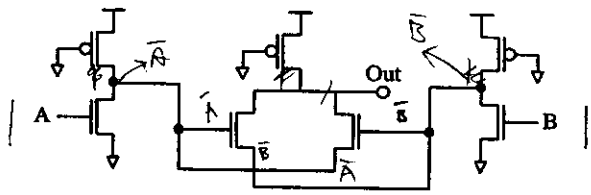


Figure 1: A logic circuit.

3. (14%) Figure 2 shows a layout view of a COMS circuit. Draw the transistor schematic diagram of the layout.

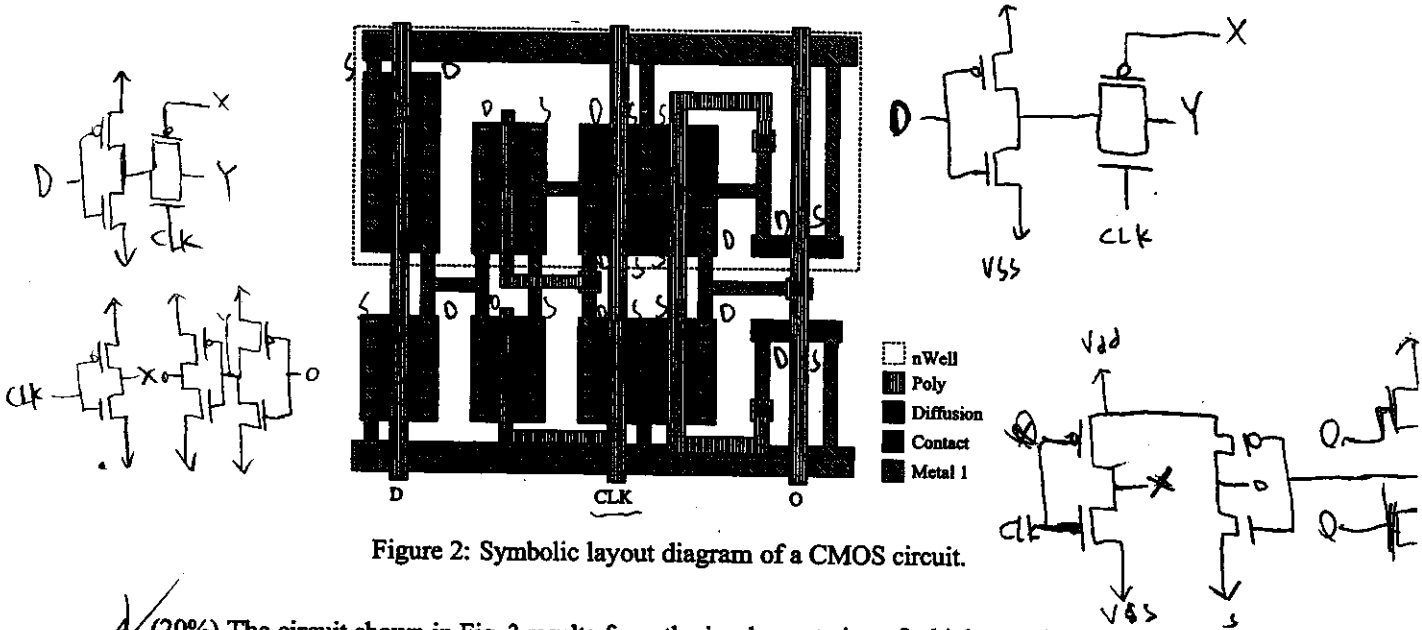


Figure 2: Symbolic layout diagram of a CMOS circuit.

4. (20%) The circuit shown in Fig. 3 results from the implementation of a high-speed one-bit full adder. Complete the following problems.
   (a) What type of logic was used to implement this circuit? (5%)

(b) Describe the operation of the circuit. (5%)

(c) Suppose that the drain capacitance and gate capacitance of the PMOS and NMOS are 0.2fF and 1fF, respectively. The leakage current of the output nodes $\overline{C}_{n+1}$ and $\overline{S}_n$ is 1nA. Also, the $V_{dd} = 1.8V$ and the switching threshold voltage of the inverter is $V_{dd}/2$. Determine that the lowest speed of the circuit should be operated (10%).

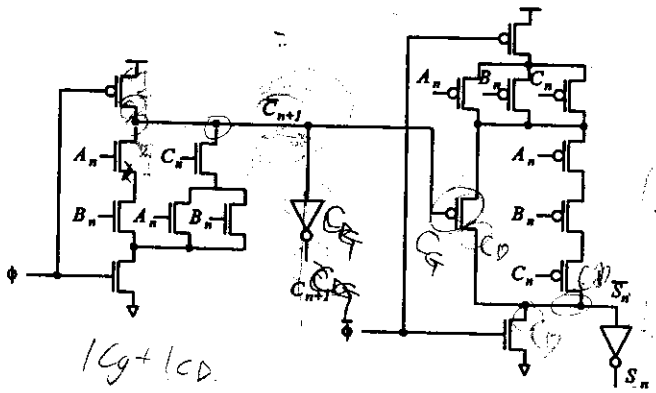


Figure 3: An adder circuit.

6. (15%) Sketch the boolean function $f = \overline{A(B+C)+DE}$ using each of the following circuit techniques. (1) Domino logic (5%) (2) Fully differential logic (5%) (3) Static CMOS logic (5%)

6. (10%) Explain the operation principle of the circuit shown in Fig. 4.

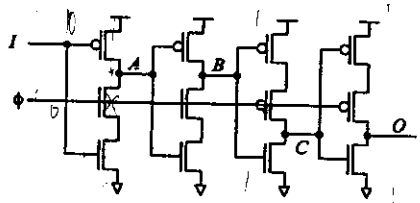


Figure 4: A CMOS circuit.

7. (10%) Fig. 5 shows a sequential circuit (pipelined circuit). Assume that the propagation delay of the combinational circuit is $15ns$. Also, the clock-to-Q delay, setup time, and hold time of the registers R1 and R2 are $2ns$, $3ns$, and $1ns$, respectively. Determine the clock period of Clk such that the circuit can run at the highest operation speed.

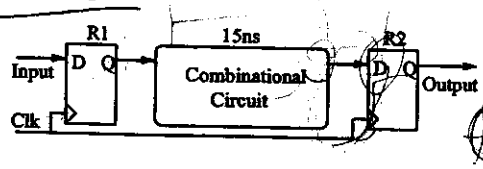


Figure 5: A pipelined circuit.